

MATH 162: Calculus II

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The Exponential Function - e^x or $\exp(x)$

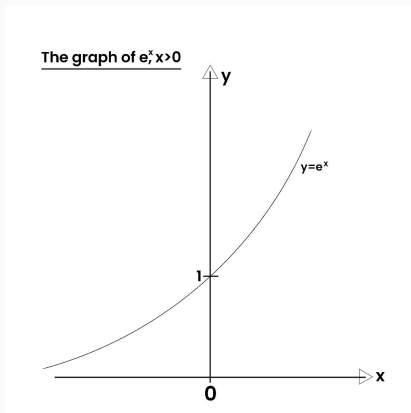
The function $y = a^x$, where $a > 0$ and x is a variable is called an exponential function. This function can be defined for integral and rational exponents as well as irrational exponents. The natural exponential function $y = e^x$, where x is a variable and $e = 2.7183$, called the Euler number is defined as an infinite series as

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

This function is often referred to as simply the exponential function.



The Graph of e^x , $x > 0$



From the graph above, $y = e^x$ is asymptotic to the x-axis and increases for all real values of x .



Properties of e^x

1. $e^0 = 1.$

2. $\lim_{x \rightarrow -\infty} e^x = 0$

3. $\lim_{x \rightarrow \infty} e^x = \infty$

4. $e^x e^y = e^{x+y}$

5. $\frac{e^x}{e^y} = e^{x-y}$

6. $(e^x)^r = e^{xr}$



Derivative of $y = e^{f(x)}$

For a differentiable function f , the derivative of $y = e^{f(x)}$ is $y'(x) = f'(x)e^{f(x)}$.

Examples

Find the derivative of each of the following.

1. $y = e^{x^2}$

2. $y = e^{3x^2}$

3. $y = e^{3x+1}$

4. $y = e^{5/x}$

5. $y = e^{\sqrt{x}}$

6. $y = e^{x \sin x}$



The Integral of $y = e^{f(x)}$

If f is a differentiable function of x such that $f'(x)$ is not a function of x then $\int e^{f(x)} dx = \frac{1}{f'(x)} e^{f(x)} + c$.

Examples

Evaluate the following.

1. $\int e^{2x+1} dx$

2. $\int e^{x/3} dx$

3. $\int x^2 e^{-x^3} dx$

4. $\int 3e^{x/2} dx$

5. $\int \frac{x}{2} e^{x^2} dx$

6. $\int x^3 e^{-x^4} dx$



The Logarithmic Function

Since logarithm is an index, the discussion of exponential functions leads naturally to the consideration of logarithms.

The logarithm of a number b to the base a , written $\log_a b$, is the power to which the base must be raised to equal b . i.e.

$$\log_a b = k \implies b = a^k.$$



The Natural Logarithm

If the base of the logarithm is e , the Euler's number, then it is called natural logarithm.

The notation

$$\log_e x$$

is commonly denoted as

$$\ln x.$$

Remark

Since $\ln x$, for $x < 0$ is not defined, $\ln |x|$ is preferred.



Properties of the Natural Logarithm

1. $\ln xy = \ln x + \ln y$
2. $\ln \frac{x}{y} = \ln x - \ln y$
3. $\ln x^a = a \ln x$, where a is a constant.
4. $\ln b = \frac{\log_a b}{\log_a e}$
5. $\ln e = 1$
6. $\lim_{x \rightarrow \infty} \ln x = \infty$
7. $\lim_{x \rightarrow 0^+} \ln x = -\infty$
8. $\ln x$ is a strictly increasing function.



Examples

1. Express each of the following as the sum or difference of logarithm.

(a) $\ln \cot x$

(b) $\ln \tan^2 x$

(c) $\ln (x^2 - 4)$

(d) $\ln \sqrt{\frac{x+1}{x-1}}$

(e) $\ln (3x^2 \sin^4 x)$

2. Express the following as a single logarithm

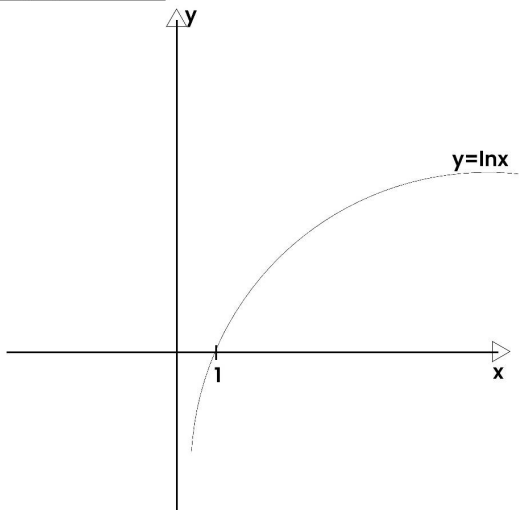
(a) $\ln x + \ln y - 1$

(b) $\ln \left(\frac{e^3}{x}\right) + \ln \frac{x}{e}$



Graph of $\ln x, x > 0$

The graph of $\ln x, x > 0$



From the graph above the following are noted.

1. $\ln x < 0 \quad \forall x \in \mathbb{R} \ni 0 < x < 1.$
2. $\ln x = 0 \quad \forall x = 1.$
3. $\ln x > 0 \quad \forall x \geq 1.$
4. $\ln x$ is asymptotic to the y-axis.



The Derivative of $\ln f(x)$

Given that

$$y = \ln(f(x)),$$

$$y' = \frac{f'(x)}{f(x)}.$$

Example

Find the derivative of each of the following functions.

1. $y = \ln(x^2 + 2)$

2. $y = \ln\left(\frac{x}{\sqrt{x^2+1}}\right)$

3. $y = \ln\left(\frac{x}{(x-1)^2}\right)$

4. $y = \ln(2 - x^3)^2$



The Derivative and the Integral of a^x , $a > 0$

If $y = a^x$, $a > 0$ then $\frac{d}{dx}(a^x)$ is obtained as follows:

$$\text{Let } y = a^x$$

$$\implies \ln y = x \ln a$$

$$\implies \frac{1}{y} \frac{dy}{dx} = \ln a$$

$$\implies \frac{dy}{dx} = y \ln a$$

$$\implies \frac{dy}{dx} = a^x \ln a$$

$$\therefore \frac{d}{dx}(a^x) = a^x \ln a \dots\dots\dots(1)$$

$$\text{From (1), } \int a^x \ln a \, dx = a^x + k$$

$$\implies \ln a \int a^x \, dx = a^x + k$$

$$\therefore \int a^x \, dx = \frac{a^x}{\ln a} + c \dots\dots\dots(2)$$



Examples

1. Find $\frac{d}{dx}(5^x)$ and deduce $\int 5^x dx$.
2. Evaluate $\frac{d}{dx}(2^{x^2})$ and deduce $\int 2^{x^2} dx$.
3. Evaluate each of the following integrals.
 - (a) $\int 3e^{2x} dx$
 - (b) $\int x^2 e^{x^3} dx$
 - (c) $\int 2^{\tan x} \sec^2 x dx$



Evaluation of Integrals of the form $\int \frac{f'(x)}{f(x)} dx$

The results of

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$

holds for all rational values of n , except $n = -1$. We explore avenues to overcome this difficulty. From $\frac{d}{dx}(\ln x) = \frac{1}{x}$ it follows that

$$\int \frac{1}{x} dx = \ln |x| + c$$

or

$$\int \frac{1}{x} dx = \ln |kx|.$$

where $c = \ln |k|$.



Example

Evaluate the following integrals.

$$1. \int \frac{1}{2x}, dx$$

$$2. \int \frac{1}{2x-1}, dx$$

$$3. \int \frac{x}{x^2+1}, dx$$

$$4. \int \frac{1}{2x+3}, dx$$

$$5. \int \frac{1}{1-x}, dx$$

$$6. \int \frac{x^2}{x^3-2}, dx$$

$$7. \int \frac{x}{x-1}, dx$$



The Definite Integral $\int_a^b \frac{dx}{x}$ when $a < 0$ and $b < 0$

The function $y = \ln x$ takes only positive values of x i.e. $\ln x$ is not defined when $x < 0$. Since

$$\int_a^b \frac{dx}{x} = \ln x \Big|_a^b$$

we are confronted with a difficulty which may be demonstrated in graphical terms as follows.



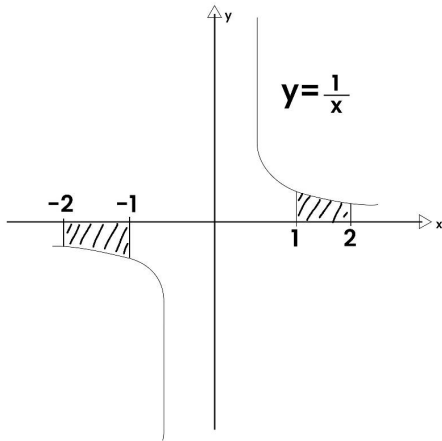


Fig 1



The integral

$$\int_{-2}^{-1} \frac{dx}{x}$$

gives the area of the curve $y = \frac{1}{x}$ bounded by the x-axis, the lines $x = -2$ and $x = -1$.

$$\begin{aligned}\text{Now, } \int_{-2}^{-1} \frac{dx}{x} &= \ln x \Big|_{-2}^{-1} \\ &= \ln(-1) - \ln(-2) \\ &= \ln\left(\frac{-1}{-2}\right) \\ &= \ln\left(\frac{1}{2}\right) \\ &= \ln(2^{-1}) \\ &= -\ln(2)\end{aligned}$$



Even though we've been able to get a value for the area, the working is not valid since $\ln(-1) - \ln(-2)$ is meaningless. We surmount this difficulty by recognising that for $x < 0$ although $\ln x$ is not defined, $\ln(-1)$ exists and $\frac{d}{dx} \ln(-x) = \frac{-1}{-x} = \frac{1}{x}$.

Then

$$\int_a^b \frac{1}{x} dx = \ln(-x) \Big]_a^b$$

This can be simplified as

$$\int_a^b \frac{1}{x} dx = \ln|x| \Big]_a^b$$



Remark

1.

$$\int_a^b \frac{1}{x} dx = \ln |x| \Big]_a^b$$

is used when a and b are both positive or are both negative.

2. If a and b have opposite signs, then

$$\int_a^b \frac{1}{x} dx$$

is not defined since $y = \frac{1}{x}$ is not continuous at $x = 0$.



Example

Evaluate the following

1. $\int_{-4}^{-3} \frac{1}{x} dx$

2. $\int_{-1}^{-1/2} \frac{1}{x} dx$

Example

Can any meaning be assigned to $\int_{-2}^2 \frac{1}{x} dx$



Integration by Parts

This technique of integration is based on the idea of differentiating the product of two functions of x . If u and v are two differentiable functions of x then,

$$\frac{d}{dx}(uv) = v \frac{du}{dx} + u \frac{dv}{dx} \dots\dots\dots(1)$$

Integrating each side of (1) with respect to x gives

$$\begin{aligned} \int \frac{d}{dx}(uv) dx &= \int v \frac{du}{dx} dx + \int u \frac{dv}{dx} dx \\ \implies uv &= \int v du + \int u dv \end{aligned}$$



$$\therefore \int u dv = uv - \int v du \dots\dots\dots(2)$$

Remark

The list below gives some of the types of combination of functions which integration by parts can apply.

- polynomial and trigonometric
- polynomial and exponential
- polynomial and logarithm
- inverse trigonometric functions
- logarithmic functions
- exponential and trigonometric
- exponential and logarithm



Examples

Evaluate each of the following integrals.

1. $\int x \cos x \, dx$

2. $\int x \sin x \, dx$

3. $\int x \cos 2x \, dx$

4. $\int x e^x \, dx$

5. $\int x \ln x \, dx$

6. $\int x^2 \sin x \, dx$

7. $\int x^2 e^x \, dx$

8. $\int x^2 \cos x \, dx$

9. $\int \ln x \, dx$

10. $\int \tan^{-1} x \, dx$

11. $\int \sin^{-1} x \, dx$

12. $\int \cos^{-1} x \, dx$



The Integral of $\ln x$

The technique of integration by parts is used to find

$$\int \ln x \, dx.$$

To do this, we let $u = \ln x$ and $dv = dx$.

$$\implies \frac{du}{dx} = \frac{1}{x} \implies du = \frac{dx}{x} \quad \text{and} \quad \int dv = \int dx$$

$$\implies u = x.$$

$$\begin{aligned} \therefore \int \ln x \, dx &= x \ln x - \int x \frac{dx}{x} \\ &= x \ln x - \int dx \\ &= x \ln x - x + c. \end{aligned}$$



Example

Evaluate each of the following integrals.

1. $\int x \ln x \, dx$

2. $\int e^x \ln x \, dx$

3. $\int x^2 \ln x \, dx$



Trigonometric Functions

The trigonometric functions are

1. $\sin x$

2. $\cos x$

3. $\tan x$

4. $\operatorname{cosec} x = \frac{1}{\sin x}$

5. $\sec x = \frac{1}{\cos x}$

6. $\cot x = \frac{1}{\tan x}$



Differentiation of Trigonometric Functions

1. $\frac{d}{dx}(\sin x) = \cos x$
2. $\frac{d}{dx}(\cos x) = -\sin x$
3. $\frac{d}{dx}(\tan x) = \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right) = \sec^2 x$
4. $\frac{d}{dx}(\sec x) = \sec x \tan x$
5. $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$
6. $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$

In general, if $y = \sin f(x)$ where $f(x)$ is a differentiable function of x then, $y'(x) = f'(x) \cos f(x)$



Example

Find the derivatives of the following trigonometric functions.

1. $y = \cos 4x$

2. $y = \sin 4x^2 + 1$

3. $y = \tan x^2$

4. $y = \cos x + \sin x$

5. $y = 3 \cos 5x^2 + 2$



Derivative of Powers of Trigonometric Functions

The differentiation of powers of trigonometric functions is made easier by using the chain rule of differentiation. If

$$y = f(u) \text{ and } u = f(x) \text{ then } \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

In general, if $y = \sin^n x$ then

$$\frac{dy}{dx} = n \sin^{n-1} x \cdot \cos x.$$

Similarly if

$$y = \cos^n x$$

then

$$\frac{dy}{dx} = n \cos^{n-1} x \cdot (-\sin x).$$



Example

Differentiate the following with respect to x .

1. $y = \sin^3 x$

2. $y = \cos^4 x$

3. $y = \tan^2 x$

4. $y = 2 \cos^2 (x^2 + 1)$

5. $y = \sin^2 (4x^3)$

6. $y = 3 \tan^4 x$

7. $y = x^4 \cos x$

8. $y = \tan^9 4x^2$



Differentiation of Inverse Trigonometric Functions

The functions $\sin^{-1} x$ ($\arcsin x$), $\cos^{-1} x$ ($\arccos x$) and $\tan^{-1} x$ ($\arctan x$) will be considered.

1. $\sin^{-1} x$:

If $y = \sin^{-1} x$ then $\sin y = x$. To find $\frac{dy}{dx}$, we use the implicit rule of differentiation.

$$\text{Let } y = \sin^{-1} x$$

$$\implies \sin y = x$$

$$\implies \cos y \frac{dy}{dx} = 1$$

$$\implies \frac{dy}{dx} = \frac{1}{\cos y}$$

$$\text{But } \sin^2 y + \cos^2 y = 1$$

$$\implies \cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$$

$$\therefore \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$



2. $\cos^{-1} x$:

Similarly, if $y = \sin^{-1} x$ then

$$\frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$$

3. $\tan^{-1} x$

Let $y = \tan^{-1} x$

$$\implies \tan y = x$$

$$\implies \sec^2 y \frac{dy}{dx} = 1$$

$$\therefore \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

But $1 + \tan^2 y = \sec^2 y$

$$\implies \frac{dy}{dx} = \frac{1}{1+\tan^2 y} = \frac{1}{1+x^2}.$$



Examples

1. If $y = \tan^{-1}(x^2 + 1)$ Show that $\frac{dy}{dx} = \frac{2x}{x^4 + 2x^2 + 2}$
2. Differentiate each of the following with respect to x .

(a) $\cos^{-1} x$

(e) $\operatorname{cosec}^{-1} x$

(b) $\sin^{-1}(2x + 1)$

(f) $\sin^{-1} x^2$

(c) $\cot^{-1} x$

(g) $\cos^{-1}(2x - 1)$

(d) $\sec^{-1} x$

3. Find each of the following derivatives.

(a) $\frac{d}{dx}(\sin^{-1} x + \cos^{-1} x)$

(b) $\frac{d}{dx}(\tan^{-1} x + \cot^{-1} x)$



Integration of Trigonometric Functions.

$$1. \int \sin x \, dx = -\cos x + c$$

$$2. \int \cos x \, dx = \sin x + c$$

$$3. \int \tan x \, dx = -\ln |\cos x| + c = \ln (\sec x) + c$$

$$4. \int \sec^2 x \, dx = \tan x + c$$

$$5. \int \operatorname{cosec}^2 x \, dx = -\cot x + c$$

$$6. \int \sec x \tan x \, dx = \sec x + c$$

$$7. \int \operatorname{cosec} x \cot x \, dx = \operatorname{cosec} x + c$$



In general,

$$\int \cos (ax + b) dx = \frac{1}{a} \sin (ax + b) + c$$

$$\int \sin (ax + b) dx = -\frac{1}{a} \cos (ax + b) + c$$

$$\int \tan (ax + b) dx = \frac{1}{a} \ln |\sec (ax + b)| + c$$



Examples

Evaluate each of the following integrals.

1. $\int 3 \sec^2 x \, dx$

2. $\int \sin 4x \, dx$

3. $\int 3 \cos (4x - \pi/2) \, dx$

4. $\int (\sec^2 2x - \operatorname{cosec}^2 4x) \, dx$

5. $\int (\cos 3x + 3 \sin x) \, dx$

6. $\int 2 \sec^2 4x \, dx$



Integration of Powers of $\sin x$, $\cos x$ and $\tan x$

1. Even Powers:

For even powers of $\sin x$ and $\cos x$, we use the identities

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

and

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$



Examples

Find each of the following integrals.

1. $\int \sin^2 x \, dx$

2. $\int \cos^2 2x \, dx$

3. $\int \sin^4 x \, dx$

4. $\int \cos^4 3x \, dx$

5. $\int \sin^2 6x \, dx$



2. Odd Powers:

For odd powers of $\sin x$ and $\cos x$, the identity

$$\sin^2 x + \cos^2 x = 1 \text{ is used.}$$

The odd power is then split into the product of an even power and an odd power of 1. For instance, $\sin^5 x$ is simplified to be

$$\sin^5 x = \sin^3 x \sin^2 x \dots\dots(1)$$

or

$$\sin^5 x = \sin^4 x \sin x \dots\dots(2)$$

but the preferred one is (2).



Examples

Integrate the following

1. $\int \sin^3 x \, dx$

2. $\int \sin^5 2x \, dx$

3. $\int \cos^3 4x \, dx$

4. $\int 3 \cos^3 (2x + 1) \, dx$

5. $\int \sin^3 6x \, dx$



3. **Powers of $\tan x$:**

For such integrals, we use the identity

$$1 + \tan^2 x = \sec^2 x$$

or

$$\tan^2 x = \sec^2 x - 1.$$



Examples

Find the following integrals

1. $\int \tan^2 x \, dx$

2. $\int \tan^3 4x \, dx$

3. $\int \tan^4 2x \, dx$

4. $\int \tan^5 x \, dx$



Integrals of the form

$$\int \sin x \cos^n x \, dx, \int \cos x \sin^n x \, dx, \int \sec^n x \tan x \, dx$$

For such integrals, the technique is to use substitution or $(n + 1)$ inspection which says that

1.

$$\int \sin x \cos^n x \, dx = -\frac{1}{n+1} \cos^{n+1} x + c$$

and

2.

$$\int \cos x \sin^n x \, dx = \frac{1}{n+1} \sin^{n+1} x + c.$$



Examples

Find the following integrals

1. $\int \sin 3x \cos^2 3x \, dx$

2. $\int \sec^2 x \tan^2 x \, dx$

3. $\int \sec^5 x \tan x \, dx$

4. $\int \cos x \sin^4 x \, dx$



Integration by Substitution

1.

$$\int (ax + b)^n dx = \frac{1}{a(n+1)}(ax + b)^{n+1}$$

where a and b are constants.

Use the substitution $u = ax + b$.

2.

$$\int f'(x)\{f(x)\}^n dx$$

or

$$\int \frac{f'(x)}{\{f(x)\}^n} dx .$$

Use the substitution $u = f(x)$.



Examples

Find the following integrals

$$1. \int \frac{3x}{\sqrt{2x^2 + 1}} dx$$

$$2. \int 5x\sqrt{2x^2 + 1} dx$$

$$3. \int \frac{x - 1}{(x + 1)^3} dx$$

$$4. \int x^2\sqrt{x^3 + 1} dx$$

$$5. \int \sin^4 4x \cos 4x dx$$

$$6. \int x(3x^2 + 2)^4 dx$$

$$7. \int x(2x + 3)^{1/2} dx$$



Integration of Products of Sine and Cosine with Linear Arguments

We now look at integrals involving products of sine and cosine functions having the same or different linear arguments. This technique reduces what may seem a very complex integration. The procedure works better if one can identify the identity to be used for each integration.



Functions with Different Arguments

These integrals are of the form

$$\int \sin \alpha x \cos \beta x \, dx,$$

or

$$\int \sin \alpha x \sin \beta x \, dx,$$

or

$$\int \cos \alpha x \cos \beta x \, dx .$$

The table below gives a summary of the identities to be used in such integrals.



	Product Identity	Source
1.	$\sin \alpha \sin \beta = \frac{1}{2} \cos (\alpha - \beta) - \frac{1}{2} \cos (\alpha + \beta)$	$\cos (\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$ $\cos (\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$
2.	$\cos \alpha \cos \beta = \frac{1}{2} \cos (\alpha - \beta) + \frac{1}{2} \cos (\alpha + \beta)$	$\cos (\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$ $\cos (\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$
3.	$\sin \alpha \cos \beta = \frac{1}{2} \sin (\alpha - \beta) + \frac{1}{2} \sin (\alpha + \beta)$	$\sin (\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$ $\sin (\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$
4.	$\cos \alpha \sin \beta = \frac{1}{2} \sin (\alpha + \beta) - \frac{1}{2} \sin (\alpha - \beta)$	$\sin (\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$ $\sin (\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$

Remark

The technique/procedure involves applying the above identities to reduce the complexities of the integrand.



Examples

Evaluate each of the following integrals.

1. $\int \sin 2x \sin 3x \, dx$

2. $\int \cos 5x \cos 9x \, dx$

3. $\int \sin 9x \cos 4x \, dx$

4. $\int \sin^2 3x \cos 7x \, dx$



Solution

$$\begin{aligned} 1. \quad & \int \sin 2x \sin 3x \, dx \\ &= \int \frac{1}{2} \cos (2x - 3x) \, dx - \int \frac{1}{2} \cos (2x + 3x) \, dx \\ &= \frac{1}{2} \int \cos (-x) \, dx - \frac{1}{2} \int \cos (5x) \, dx \\ &= -\frac{1}{2} \sin (-x) - \frac{1}{10} \sin 5x + c \\ \\ &\therefore \int \sin 2x \sin 3x \, dx = \frac{1}{2} \sin (x) - \frac{1}{10} \sin 5x + c \end{aligned}$$



$$\begin{aligned}
 2. \quad & \int \cos 5x \cos 9x \, dx \\
 &= \int \frac{1}{2} \cos (5x - 9x) \, dx + \int \frac{1}{2} \cos (5x + 9x) \, dx \\
 &= \frac{1}{2} \int \cos (-4x) \, dx + \frac{1}{2} \int \cos (14x) \, dx \\
 &= \frac{1}{8} \sin 4x + \frac{1}{28} \sin 4x + c
 \end{aligned}$$

$$\therefore \int \cos 5x \cos 9x \, dx = \frac{1}{8} \sin (4x) + \frac{1}{28} \sin 14x + c$$



$$\begin{aligned}
 3. \quad & \int \sin 9x \cos 4x \, dx \\
 &= \int \frac{1}{2} \sin (9x - 4x) \, dx + \int \frac{1}{2} \sin (9x + 4x) \, dx \\
 &= \frac{1}{2} \int \sin (5x) \, dx + \frac{1}{2} \int \sin (13x) \, dx \\
 &= -\frac{1}{10} \cos 5x - \frac{1}{26} \cos 13x + c
 \end{aligned}$$

$$\therefore \int \sin 9x \cos 4x \, dx = -\frac{1}{10} \cos (5x) - \frac{1}{26} \cos 13x + c$$



$$\begin{aligned}
 4. \quad & \int \sin^2 3x \cos 7x \, dx \\
 &= \int \frac{1}{2}(1 - \cos 6x) \cos 7x \, dx \\
 &= \frac{1}{2} \int \cos(7x) \, dx - \frac{1}{2} \int \cos 6x \cos 7x \, dx \\
 &= \frac{1}{2} \int \cos(7x) \, dx - \frac{1}{2} \int \frac{1}{2} \cos x \, dx - \frac{1}{2} \int \frac{1}{2} \cos 13x \, dx \\
 &= \frac{1}{14} \sin 7x - \frac{1}{4} \sin x - \frac{1}{52} \sin 13x + c
 \end{aligned}$$

$$\therefore \int \sin^2 3x \cos 7x \, dx = \frac{1}{14} \sin 7x - \frac{1}{4} \sin x - \frac{1}{52} \sin 13x + c$$



Examples

Evaluate each of the following integrals.

1. $\int \sin 2x \cos 3x \, dx$

2. $\int \sin 4x \sin 3x \, dx$

3. $\int \cos 3x \cos 4x \, dx$

4. $\int \cos 2x \cos 3x \, dx$

5. $\int \cos 2x \sin 3x \, dx$

6. $\int \sin 2x \sin 3x \, dx$

7. $\int_0^{\frac{\pi}{2}} 4 \sin x \cos \frac{1}{3}x \, dx$

8. $\int_0^{\frac{\pi}{2}} 4 \sin \frac{1}{3}x \cos x \, dx$

9. $\int \sin 6x \sin 2x \, dx$

10. $\int \cos 5x \cos^2 2x \, dx$

11. $\int \sin^3 4x \cos 5x \, dx$



Functions with Same Arguments

	Product Identity	Source
1.	$\sin \alpha \cos \alpha = \frac{1}{2} \sin 2\alpha$	$\sin 2\alpha = \sin (\alpha + \alpha)$ $= \sin \alpha \cos \alpha + \cos \alpha \sin \alpha$ $= 2 \sin \alpha \cos \alpha$
2.	$\cos \alpha \cos \alpha = \cos^2 \alpha = \frac{1}{2}(1 + \cos 2\alpha)$	$\cos 2\alpha = \cos (\alpha + \alpha) = \cos^2 \alpha - \sin^2 \alpha$ $= \cos^2 \alpha - (1 - \cos^2 \alpha)$ $= 2 \cos^2 \alpha - 1$
3.	$\sin \alpha \sin \alpha = \sin^2 \alpha = \frac{1}{2}(1 - \cos 2\alpha)$	$\cos 2\alpha = \cos (\alpha + \alpha) = \cos^2 \alpha - \sin^2 \alpha$ $= 1 - \sin^2 \alpha - \sin^2 \alpha$ $= 1 - 2 \sin^2 \alpha$



Integration Using Inverse Trigonometric Functions

Integrals of the form

$$1. \int \frac{1}{\sqrt{a^2 - b^2 x^2}} dx \quad \text{or}$$

$$2. \int \frac{1}{a^2 + b^2 x^2} dx$$

require trigonometric substitution. The idea is to write

$$a^2 - b^2 x^2 \quad \text{as} \quad a^2 - a^2 \sin^2 \theta = a^2(1 - \sin^2 \theta) = a^2 \cos^2 \theta$$

or

$$a^2 + b^2 x^2 \quad \text{as} \quad a^2 + a^2 \tan^2 \theta = a^2(1 + \tan^2 \theta) = a^2 \sec^2 \theta.$$



To achieve this, we use the substitution

$$x = \frac{a}{b} \sin \theta$$

for

$$\int \frac{1}{\sqrt{a^2 - b^2 x^2}} dx$$

and

$$x = \frac{a}{b} \tan \theta$$

for

$$\int \frac{1}{a^2 + b^2 x^2} dx .$$



These substitutions then give

$$dx = \frac{a}{b} \cos \theta \, d\theta$$

and

$$dx = \frac{a}{b} \sec^2 \theta \, d\theta.$$

Therefore,

$$\int \frac{1}{\sqrt{a^2 - b^2 x^2}} dx = \int \frac{\frac{a}{b} \cos \theta}{\sqrt{a^2 \cos^2 \theta}} d\theta = \frac{1}{b} \int \frac{a \cos \theta}{a \cos \theta} d\theta,$$

from which

$$\int \frac{1}{\sqrt{a^2 - b^2 x^2}} dx = \frac{1}{b} \sin^{-1} \frac{b}{a} x + c.$$



Finally, the integral

$$\int \frac{1}{a^2 + b^2 x^2} dx$$

gives

$$\int \frac{1}{a^2 + b^2 x^2} dx = \int \frac{\frac{a}{b} \sec^2 \theta}{a^2 \sec^2 \theta} d\theta = \frac{1}{b} \int \frac{a \sec^2 \theta}{a^2 \sec^2 \theta} d\theta,$$

giving

$$\int \frac{1}{a^2 + b^2 x^2} dx = \frac{1}{b} \tan^{-1} \frac{b}{a} x + c.$$



Remark

$$\int \frac{1}{\sqrt{a^2 - b^2 x^2}} dx = -\frac{1}{b} \cos^{-1} \frac{b}{a} x + c,$$

when the substitution $x = \frac{a}{b} \cos \theta$ is used.



Examples

Find the following integrals

$$1. \int \frac{1}{\sqrt{1-x^2}} dx$$

$$2. \int \frac{1}{1+x^2} dx$$

$$3. \int \frac{1}{\sqrt{9-4x^2}} dx$$

$$4. \int \frac{1}{\sqrt{4-x^2}} dx$$

$$5. \int \frac{1}{\sqrt{1-3x^2}} dx$$

$$6. \int \frac{1}{\sqrt{16-9x^2}} dx$$

$$7. \int \frac{1}{4+x^2} dx$$

$$8. \int \frac{1}{1+16x^2} dx$$

$$9. \int \frac{1}{3+4x^2} dx$$

$$10. \int \frac{2}{1+x^2} dx$$

$$11. \int \frac{3}{\sqrt{1-x^2}} dx$$

$$12. \int \frac{1}{\sqrt{25-x^2}} dx$$

$$13. \int \frac{3}{4+x^2} dx$$

$$14. \int \frac{-4}{16-4x^2} dx$$



Hyperbolic Functions

The hyperbolic functions are

- hyperbolic sine also called $\sinh$.
- hyperbolic cosine also called $\cosh$.
- hyperbolic tan also called $\tanh$.



Identities of Hyperbolic Functions

$$1. \sinh(x) = \frac{e^x - e^{-x}}{2}$$

$$2. \cosh(x) = \frac{e^x + e^{-x}}{2}$$

$$3. \tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{\frac{e^x - e^{-x}}{2}}{\frac{e^x + e^{-x}}{2}} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$4. \operatorname{sech}(x) = \frac{1}{\cosh(x)} = \frac{2}{e^x + e^{-x}}$$

$$5. \operatorname{cosech}(x) = \frac{1}{\sinh(x)} = \frac{2}{e^x - e^{-x}}$$

$$6. \coth(x) = \frac{1}{\tanh(x)} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$



Osborn's Rule

Osborn's rule is a rule for converting a trigonometric identity into a corresponding hyperbolic identity.

It provides a simple way to remember the change of signs between trigonometric identities and hyperbolic identities.

There are obvious similarities between the identities connecting hyperbolic functions and the corresponding trigonometric functions. The only difference between these two sets of identities is only a matter of sign.



The rule states that every occurrence of a sine or a cosine can be replaced with the corresponding hyperbolic sine or hyperbolic cosine, and wherever there exists a product of two sines, the product of the hyperbolic sine must be negated.



Remark

Since all these ratios contain a sine, their products also contain the product of two sines.

$$1. \tan x = \frac{\sin x}{\cos x}$$

$$2. \cot x = \frac{\cos x}{\sin x}$$

$$3. \operatorname{cosec} x = \frac{1}{\sin x}$$

For instance,

- $\sin 2x = 2 \sin x \cos x \implies \sinh 2x = 2 \sinh x \cosh x$
- $\cos 2x = \cos^2 x - \sin^2 x \implies \cosh 2x = \cosh^2 x + \sinh^2 x$
- $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x} \implies \tanh 2x = \frac{2 \tanh x}{1 + \tanh^2 x}$



Trial Exercise

Find the corresponding hyperbolic identity for each of the following trigonometric identities.

1. $\sin \alpha + \sin \beta = 2 \sin \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta)$

2. $\cos \alpha + \cos \beta = 2 \cos \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta)$

3. $\cos \alpha - \cos \beta = -2 \sin \frac{1}{2}(\alpha + \beta) \sin \frac{1}{2}(\alpha - \beta)$

4. $\sin^2 \alpha + \cos^2 \alpha = 1$

5. $\sec^2 \alpha = 1 + \tan^2 \alpha$

6. $\operatorname{cosec}^2 \alpha = 1 + \cot^2 \alpha$

7. $\cos 3\alpha = 4 \cos^4 \alpha - 3 \cos \alpha$

8. $\tan 3\alpha = \frac{3 \tan \alpha - \tan^3 \alpha}{1 - 3 \tan^2 \alpha}$



Example

1. Prove each of the following identities

(a) $\cosh x - \sinh^2 x = 1$

(b) $1 - \tanh^2 x = \operatorname{sech}^2 x$

(c) $1 - \coth^2 x = \operatorname{cosech}^2 x$

(d) $\cosh x + \sinh x = e^x$

(e) $\cosh x - \sinh x = e^{-x}$

2. Use the definition $\sinh x = \frac{1}{2}(e^x - e^{-x})$ and $\cosh x = \frac{1}{2}(e^x + e^{-x})$ to show that

(a) $\cosh(-x) = \cosh x$

(b) $\sinh(-x) = -\sinh x$

(c) $\cosh x > \sinh x$



Derivatives of Hyperbolic Functions

$$\begin{aligned} 1. \quad \frac{d}{dx}(\cosh x) &= \frac{d}{dx}\left(\frac{1}{2}(e^x + e^{-x})\right) \\ &= \frac{1}{2}(e^x - e^{-x}) \\ &= \sinh x \end{aligned}$$

$$\therefore \frac{d}{dx}(\cosh x) = \sinh x$$

$$\begin{aligned} 2. \quad \frac{d}{dx}(\sinh x) &= \frac{d}{dx}\left(\frac{1}{2}(e^x - e^{-x})\right) \\ &= \frac{1}{2}(e^x + e^{-x}) \\ &= \cosh x \end{aligned}$$

$$\therefore \frac{d}{dx}(\sinh x) = \cosh x$$



Remark

The derivatives of other hyperbolic functions are easily obtained by first expressing them in terms of $\cosh x$ and $\sinh x$.



Inverse Hyperbolic Functions

The inverse of $\cosh x$, $\sinh x$ and $\tanh x$ are expressed in terms of logarithm.

1. $\sinh^{-1} x$

Let $y = \sinh^{-1} x$

$$\implies \sinh y = x$$

But $\cosh^2 y - \sinh^2 y = 1$

$$\implies \cosh^2 y = 1 + \sinh^2 y$$

$$\implies \cosh^2 y = 1 + x^2$$

$$\therefore \cosh y = \sqrt{1 + x^2}$$

Now, $\sinh y + \cosh y = x + \sqrt{1 + x^2} \dots\dots\dots(1)$

Also, $\sinh y + \cosh y = \frac{1}{2}(e^y - e^{-y}) + \frac{1}{2}(e^y + e^{-y})$

$$\implies \sinh y + \cosh y = e^y \dots\dots\dots(2)$$



From (1) and (2),

$$e^y = x + \sqrt{1 + x^2}$$

$$\therefore y = \ln \{x + \sqrt{1 + x^2}\}$$

$$\therefore \sinh^{-1} x = \ln \{x + \sqrt{1 + x^2}\}.$$



2. $\cosh^{-1} x$

Let $y = \cosh^{-1} x$

$$\implies \cosh y = x$$

But $\cosh^2 y - \sinh^2 y = 1$

$$\implies \sinh^2 y = \cosh^2 y - 1 \implies \sinh^2 y = x^2 - 1$$

$$\therefore \sinh y = \sqrt{x^2 - 1}$$

$$\text{Now, } \cosh y + \sinh y = x + \sqrt{x^2 - 1} \dots\dots\dots(1)$$

$$\text{Also, } \cosh y - \sinh y = \frac{1}{2}(e^y + e^{-y}) - \frac{1}{2}(e^y - e^{-y})$$

$$\implies \cosh y - \sinh y = e^{-y} \dots\dots\dots(2)$$



From (1) and (2),

$$e^y = x + \sqrt{x^2 - 1}$$

$$\therefore y = \ln \{x + \sqrt{x^2 - 1}\}$$

$$\therefore \cosh^{-1} x = \ln \{x + \sqrt{x^2 - 1}\}$$



3. $\tanh^{-1} x$

$$\text{Let } y = \tanh^{-1} x$$

$$\implies \tanh y = x$$

$$\implies x = \frac{\sinh y}{\cosh y} = \frac{\frac{1}{2}(e^y - e^{-y})}{\frac{1}{2}(e^y + e^{-y})} = \frac{e^y - e^{-y}}{e^y + e^{-y}}$$

$$\implies x = \frac{e^{-y}(e^{2y} - 1)}{e^{-y}(e^{2y} + 1)}$$

$$\implies x = \frac{e^{2y} - 1}{e^{2y} + 1}$$

$$\implies x(e^{2y} + 1) = e^{2y} - 1$$

$$\implies e^{2y}(1 - x) = (x + 1)$$

$$\implies e^{2y} = \frac{x+1}{1-x}$$

$$\therefore e^{2y} = \frac{x+1}{1-x}$$

$$\implies 2y = \ln \left(\frac{1+x}{1-x} \right)$$

$$\implies y = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

$$\therefore \tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$



Integration using Inverse Hyperbolic Functions

The purpose of including hyperbolic functions is to extend our ability of integration to functions which hitherto could not be integrated. So far we have considered the following.

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \frac{1}{a} \sin^{-1} \frac{x}{a} + c,$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + c$$

and

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = -\frac{1}{a} \cos^{-1} \frac{x}{a} + c.$$



How about the following integrals.

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx$$

$$\int \frac{1}{\sqrt{a^2 + x^2}} dx$$

$$\int \frac{1}{a^2 - x^2} dx$$



The identity $\sin^2 \theta + \cos^2 \theta = 1$ helps us to eliminate the square root sign in

$$\frac{1}{\sqrt{a^2 - x^2}}.$$

Similarly, the square root sign in

$$\frac{1}{\sqrt{x^2 - a^2}} \quad \text{and} \quad \frac{1}{\sqrt{x^2 + a^2}}$$

can be eliminated using the identity

$$\cosh^2 \theta - \sinh^2 \theta = 1.$$

To eliminate the difficulty in the integral $\int \frac{1}{a^2 - x^2} dx$, we use the identity $\tanh^2 \theta + \operatorname{sech}^2 \theta = 1$.



Examples

1. Show that $\int \frac{1}{\sqrt{x^2+a^2}} dx = \sinh^{-1} \frac{x}{a} + c$

2. Prove that $\int \frac{1}{\sqrt{x^2-a^2}} dx = \cosh^{-1} \frac{x}{a} + c$

3. Consider the integral $\int \frac{1}{a^2-x^2} dx$.

(a) Use partial fractions to show that

$$\int \frac{1}{a^2-x^2} dx = \frac{1}{2a} \ln \left(\frac{a+x}{a-x} \right) + c$$

(b) Use the substitution $x = a \tanh \theta$ to show that

$$\int \frac{1}{a^2-x^2} dx = \frac{1}{a} \tanh^{-1} \frac{x}{a} + c.$$



Remark

In summary the following integrals and the appropriate substitutions for each are worthy of note.

$$1. \int \frac{1}{\sqrt{a^2 - b^2 x^2}} dx, \quad \text{use } x = \frac{a}{b} \sin \theta \quad \text{or} \quad x = \frac{a}{b} \cos \theta.$$

$$2. \int \frac{1}{\sqrt{b^2 x^2 - a^2}} dx, \quad \text{use } x = \frac{a}{b} \cosh \theta.$$

$$3. \int \frac{1}{\sqrt{a^2 + b^2 x^2}} dx, \quad \text{use } x = \frac{a}{b} \sinh \theta.$$

$$4. \int \frac{1}{a^2 + b^2 x^2} dx, \quad \text{use } x = \frac{a}{b} \tan \theta.$$

$$5. \int \frac{1}{a^2 - b^2 x^2} dx, \quad \text{use } x = \frac{a}{b} \tanh \theta.$$



Examples

1. Find

$$(a) \int \frac{1}{\sqrt{x^2+2x+10}} dx$$

$$(b) \int \frac{1}{\sqrt{9+x^2}} dx$$

$$(c) \int \frac{1}{\sqrt{4+x^2}} dx$$

$$(d) \int \frac{1}{x^2+x+1} dx$$

$$(e) \int \frac{1}{\sqrt{x^2-6x+10}} dx$$

$$(f) \int \frac{1}{\sqrt{x^2+4}} dx$$

2. Use the substitution $e^x = \tan \theta$ to evaluate the following

$$(a) \int \operatorname{sech} x dx$$

$$(b) \int \cosh x dx$$

$$(a) \int \tanh x dx$$



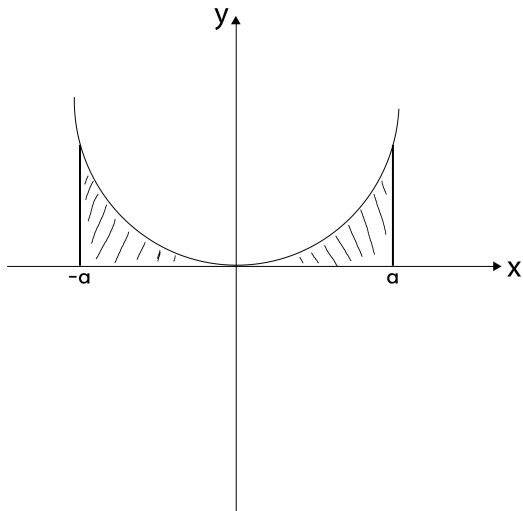
Definite Integrals of Even and Odd Functions of the form $\int_{-a}^a f(x) dx, \quad a > 0$

1. Even Functions:

A real-valued function f is said to be even if and only if $f(-x) = f(x)$ for all x in the domain of f . For instance the following functions are even.

1. $f(x) = |x|$
2. $f(x) = x^2$
3. $f(x) = \cos x$
4. $f(x) = x^{2n}$, where $n \in \mathbb{N}$





The graph of an even function is symmetric about the y-axis
i.e. the y-axis reflects the graph of such a function.

Thus the graph of even functions remain unchanged after reflection about the y-axis.

Now for any $a > 0$,

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$



Proof

Since f is even $f(-x) = f(x)$

$$\begin{aligned}\text{Thus, } \int_{-a}^a f(x) dx &= \int_{-a}^0 f(-x) dx + \int_0^a f(x) dx \\ &= \int_{-a}^0 f(x) dx + \int_0^a f(x) dx \\ &= \int_0^a f(x) dx + \int_0^a f(x) dx \\ \therefore \int_{-a}^a f(x) dx &= 2 \int_0^a f(x) dx \text{ as required.}\end{aligned}$$

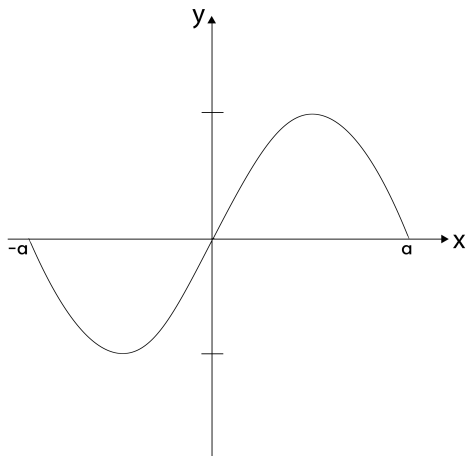


2. Odd Functions

A real-valued function f is called an odd function if for any x in the domain of f , $f(-x) = -f(x)$.

For example, the functions $f(x) = x^3$, $f(x) = x^5$,
 $f(x) = \sin x$, $f(x) = \tan x$ and $f(x) = x^{2n+1}$, $n \in \mathbb{N}$ are all odd functions.





The graph of an odd function is symmetric about the origin.

Thus, odd functions have rotational symmetry i.e. the graph of odd functions remain unchanged after rotation of 180° about the origin.

Now, for any $a > 0$,

$$\int_{-a}^a f(x) dx = 0, \text{ when } f \text{ is odd.}$$



Proof

Since f is odd $f(-x) = -f(x)$

$$\text{Thus, } \int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$$

Now for all $x \in (-a, 0)$, $f(x) = f(-x) = -f(x)$

$$\therefore \int_{-a}^a f(x) dx = - \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$$

$$\int_{-a}^a f(x) dx = - \int_0^a f(x) dx + \int_0^a f(x) dx$$

$$\therefore \int_{-a}^a f(x) dx = 0 \text{ as required.}$$



Trial Exercise

Determine whether each of the following functions is even, odd or neither even nor odd.

1. $f(x) = -5x^4 + 7x^2 - 2$

2. $f(x) = 2x^5 - 4x + 5$

3. $f(x) = \frac{3x}{x^2+1}$

4. $f(x) = \frac{x}{x^2-1}$

5. $f(x) = x^2 - 1$

6. $f(x) = |x| - x$

7. $f(x) = \tan x^2$



Ordinary Differential Equations

An equation containing any differential coefficient such as

$$\frac{dy}{dx}, \frac{d^2y}{dx^2}$$

is called an ordinary differential equation. A solution of such an equation is an equation relating x and y and containing no differential coefficient. The order of a differential equation is determined by the highest differential coefficient present in the equation.



Remark

1. The general solution of a differential equation defines some property common to a family of curves.
2. The particular solution of a differential equation is the equation of one member of the family.
3. The particular solution of a differential equation is obtained when initial conditions are given to determine any constant that may be in the general solution.



First Order Differential Equations

1. Separating the Variables

Examples

$$1. \frac{dy}{dx} = xy$$

$$2. x^2 \frac{dy}{dx} = y(y - 1)$$

$$3. \frac{dy}{dx} = \frac{x}{y}$$

$$4. \frac{dy}{dx} = \frac{y}{x}$$

$$5. \frac{dx}{dy} = xy$$

$$6. x \frac{dy}{dx} = \tan y$$

$$7. e^{-x} \frac{dy}{dx} = y^2 - 1$$

$$8. \sqrt{1 + x^2} \frac{dy}{dx} = \frac{x}{y}$$



2. Exact Equations

If it is not possible to separate the variables then we do inspect the equation to see if one side of the equation can be written as the derivative of a certain function.

Once this is possible then the equation may be solved by integrating both sides with respect to x . Such an equation is called an exact differential equation.

If not, then we find a function called an integrating factor and multiply by both sides of the equation by the integrating factor to change it to an exact equation.



Example

Solve the following equations

1. $2xy \frac{dy}{dx} + y^2 = e^2 x$

2. $x^2 \frac{dy}{dx} + 2xy = 1$

3. $x^2 \cos y \frac{dy}{dx} + 2x \sin y = \frac{1}{x}$

4. $e^y + xe^y \frac{dy}{dx} = 2$



Integrating Factors

Some differential equations that are not exact may be made so by multiplying each side by an integrating factor.

An integrating factor is a function which when multiplied by both sides of a differential equation makes the equation exact, and thus easy to integrate.



Examples

1. Solve the differential equation $xy \frac{dy}{dx} + y^2 = 3x$.
2. Find an integrating factor required to make the following differential equations exact. Hence solve it.

(a) $x \frac{dy}{dx} + 2y = e^{x^2}$

(b) $xe^y \frac{dy}{dx} + 2e^y = x$

(c) $2x^2y \frac{dy}{dx} + x^2y = 1$



3. First Order Linear Equations

A first order differential equation of the form

$$\frac{dy}{dx} + ay = b,$$

where a and b are either functions of x or constants, is said to be linear if the following conditions are satisfied.

1. No product of x and its derivative exist in the equation.
2. The degree of y and its derivative $\frac{dy}{dx}$ is 1.



Derivation of an Integrating Factor

Suppose that the equation

$$\frac{dy}{dx} + ay = b$$

can be made exact by multiplying through by an integrating factor I which is a function of x . Then

$$I \frac{dy}{dx} + Iay = bI$$

is an exact equation.

$$\implies I \frac{dy}{dx} + Iay = \frac{d}{dx}(Iy)$$

$$\implies I \frac{dy}{dx} + Iay = I \frac{dy}{dx} + y \frac{dI}{dx}$$



$$\implies y \frac{dI}{dx} = I a y$$

$$\implies \frac{dI}{dx} = a I$$

$$\implies \frac{dI}{I} = a dx$$

$$\implies \int \frac{dI}{I} dx = \int a dx$$

$$\implies \ln I(x) = \int a dx$$

$$\implies I(x) = e^{\int a dx} \dots\dots\dots (1)$$

Therefore the integrating factor required is given by (1).



Examples

1. Solve the differential equation $\frac{dy}{dx} + 3y = e^{2x}$; $y(0) = \frac{6}{5}$.
2. Solve $\frac{dy}{dx} + y \cot x = \cos x$.
3. Find the general solution of $\frac{dy}{dx} + 2xy = x$. Hence determine the particular solution with the initial condition $y(0) = -\frac{1}{2}$.
4. Solve $x \frac{dy}{dx} + 2y = 2$.



Second Order Linear Differential Equations with Constant Coefficients

Differential equations of the form

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0 \dots\dots\dots (1),$$

where a , b and c are constants, y and its derivatives are of first degree are called second-order linear differential equations with constant coefficient. To solve such an equation, we let

$$y = e^{\lambda x},$$

where λ is a constant be the assumed solution.



We then substitute $\frac{d^2y}{dx^2}$, $\frac{dy}{dx}$ and y into (1) to obtain the auxiliary quadratic equation $a\lambda^2 + b\lambda + c = 0$ (2). The roots of (2) are

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

There are three possible cases of the roots of (2).



1. If $b^2 - 4ac = 0$, then (2) has two identical real roots.
2. If $b^2 - 4ac > 0$, then (2) has two real distinct roots.
3. If $b^2 - 4ac < 0$, then (2) has one pair of conjugate complex roots.

Each case gives rise to a distinct type of solution to the differential equation (1), and we shall consider each of these cases.



Case 1: 2 Identical Real Roots

Examples

1. Find the general solution of the differential equation $4y'' - 12y' + 9y = 0$.
2. Solve the following differential equations.
 - (a) $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 0$.
 - (b) $100y'' - 60y' + 9y = 0$.
 - (c) $y'' - 2y' + y = 0$; $y(0) = 0$ and $y(1) = e$



Case 2: 2 Distinct Real Roots

Examples

Find the general solution of the following differential equations.

1. $4y'' + 2y' - 15y = 0.$

2. $y'' = y.$

3. $y'' - 12y' + 20y = 0.$

4. $15y'' - 8y' + y = 0.$

5. $2y'' - 5y' - 3y = 0.$

6. $y'' - 25y = 0.$

7. $6y'' - y' - y = 0.$

8. $y'' - 6y' + 5y = 0; \quad y(0) = 1 \text{ and } y'(0) = 9$



Case 3: Complex Conjugate Roots

Example

Solve the following differential equations.

1. $y'' + 6y' + 10y = 0.$

2. $y'' - 2y' + 50y = 0.$

3. $y'' + 6y' + 34y = 0.$

4. $36y'' + y = 0.$

5. $y'' - 4y = 0.$



The table below summarizes the three cases.

Case.	Nature of Roots of auxiliary equation	General Solution of the Differential Equation
1.	Real, identical roots ($\lambda = \lambda_1$)	$y = e^{\lambda_1 x}(Ax + B)$
2.	Real, Distinct Roots ($\lambda = \lambda_1$ or λ_2)	$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$
3.	Complex Roots ($\lambda = \alpha + \beta i$)	$y = e^{\alpha x}(A \cos \beta x + B \sin \beta x)$



The Differential Equation $ay'' + by' + cy = f(x)$

In the preceding section we have found various methods for solving any differential equation of the form

$$ay'' + by' + cy = 0 \dots\dots\dots(1).$$

We now consider the more general equation

$$ay'' + by' + cy = f(x) \dots\dots\dots(2).$$

The right hand side of (2) can be any of the following.

1. $f(x)$ is a polynomial function.
2. $f(x)$ is an exponential function.
3. $f(x)$ is a trigonometric function.



The general solution of (2) consists of two parts:

1. The solution of the equation formed by replacing $f(x)$ by zero. This solution is called the complementary function/solution.
2. An extra term required to produce $f(x)$. This part is called the particular integral/solution. i.e.

$$y_n = y_c + y_p,$$

where y_n = the general solution, y_c = the complementary solution and y_p = the particular solution.



Case 1: $f(x)$ is a polynomial.

Example

Solve the following differential equation

1. $y'' - 4y' + 3y = 9x + 6.$

2. $y'' + 4y' + 5y = 10x^2 + x.$

3. $y'' - 7y' + 12y = 4.$

4. $y'' - 4y' - 5y = 17 + 15x.$

5. $y'' - 6y' + 10y = 10x^2 + 18x - 6.$

6. $y'' + y' = 10.$



Case 2: $f(x)$ is an exponential function.

Example

Solve the following differential equation

1. $y'' + 4y' + 5y = 10e^{-2x}$.

2. $y'' + y = 10e^{2x}$.

3. $y'' + 9y = 20e^x$.

4. $16y'' - 8y' + y = 27e^x$.

5. $y'' + 2y' + y = 16e^{-x}$.



Case 3: $f(x)$ is an trigonometric function.

Example

Solve the following differential equation

1. $y'' - 4y' + 3y = 10 \sin 2x + 15 \cos 2x.$

2. $y'' + 6y' + 9y = 8 \sin 2x + 6 \cos 2x.$

3. $y'' + 3y' = 12 + \sin x.$



Some Geometrical Application of Calculus

1. Area of a Sector

Area of a sector

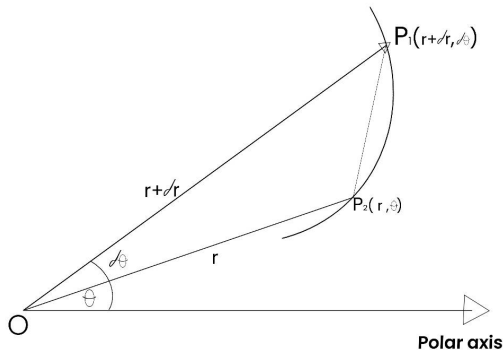


Fig 1



Let area of sector $OP_1P_2 = A_1$ and let A_2 be the area of $\triangle OP_1P_2$. Since $\partial\theta$ is small, $A_1 \simeq A_2$.

$$\text{Now, } A_1 = \frac{1}{2} P_1 P_2 r.$$

$$\text{But } \sin \partial\theta = \frac{PP_1}{r + \partial r} \implies PP_1 = (r + \partial r) \sin \partial\theta.$$

$$\therefore A_1 = \frac{1}{2} r(r + \partial r) \sin \partial\theta \dots\dots\dots (1)$$

Also, for small $\partial\theta$, $\sin \partial\theta \simeq \partial\theta$ and $r + \partial r \simeq r$ since ∂r is so small.

$$\therefore A_1 = \frac{1}{2} r^2 \partial\theta.$$

Therefore area of a sector is given by $A_1 = \int_{\alpha}^{\beta} \frac{1}{2} r^2 \partial\theta$, where α and β are values of θ corresponding to the bounding radius vectors of the sector.



2. Length of a Curve

Length of a curve

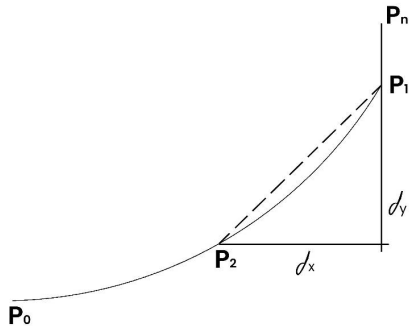


Fig 2



By Pythagoras theorem

$$|P_1P_2|^2 = (\partial x)^2 + (\partial y)^2$$

$$\Rightarrow |P_1P_2|^2 = \left\{ 1 + \left(\frac{\partial y}{\partial x} \right)^2 \right\} (\partial x)^2$$

$$\Rightarrow |P_1P_2| = \sqrt{\left\{ 1 + \left(\frac{\partial y}{\partial x} \right)^2 \right\} (\partial x)}$$

$\therefore$, Length of arc, $L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx$, where a and b are the values of x corresponding to the ends of the arc.



Remark

1. If the parametric equation of the curve are given in the form $x = f(t)$ and $y = g(t)$, then the length of the arc L is

$$L = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

2. For polar equation,

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$



Example

Find the length of the given curve between the specified points or values of the parameter.

1. $y = x^3$; $A(0, 0)$ and $B(2, 8)$
2. $y = \sqrt{x}$; $A(0, 0)$ and $B(4, 2)$
3. $r = 5 \cos \theta$; $\theta_1 = -\frac{\pi}{2}$ and $\theta_2 = -\frac{\pi}{2}$
4. $x = e^t \cos t, y = e^t \sin t$; $t_1 = 0$ and $t_2 = 2\pi$
5. $x = 5 \cos t, y = 5 \sin t$; $t_1 = 0$ and $t_2 = 2\pi$
6. $x = at^2, y = 2at$; $t_1 = 0$ and $t_2 = 5$
7. $x = at^2, y = at^3$; $t_1 = 0$ and $t_2 = 4$

